What a Tooth-Level Score Cannot Tell You: Constant Baselines, External Validation and Domain Shift in Automated Panoramic Dental Screening

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***Abstract*—Tooth decay (dental caries) and gum disease that destroys the bone around teeth (periodontal bone loss) are routinely screened for on panoramic dental radiographs, and a growing number of artificial-intelligence systems are reported to detect them automatically. Such systems are usually scored by asking which teeth they mark as diseased. We show that this score is difficult to interpret, because decay is concentrated in the back teeth: a program that ignores the radiograph entirely and always names the same eight molars can score well. We built a complete six-stage pipeline that finds each tooth, numbers it, and classifies it for decay, with a fourth stage that flags teeth near an expert-marked abnormality, and we compared it against progressively less informed systems—two general-purpose vision–language models and, finally, that image-blind constant—using one scorer on one set of 164 radiographs. The constant (F1 0.320) outscored both language models. Our pipeline cleared it on that development set—narrowly when its caries stage judged teeth one at a time (0.362), and by more under a second metric that requires the flagged tooth to sit where the lesion actually is (0.802 versus 0.630), while flagging 46 % more teeth—and its tooth detector had been trained on the same corpus. Neither lead survived outside it. On 337 radiographs from four hospitals that contributed no training data, the comparison produced no separation at all, and the difference between the two systems swung 31 percentage points across sites; on a second external benchmark of 2,815 lesions the constant won outright, by 22 percentage points. Both language models returned confident, correctly formatted tooth lists for an image containing pure random noise. The fourth stage's first score, 0.923 macro-F1, came from a split that let the same radiographs into training and validation; split by radiograph, three training runs found 11.0–27.5 % of the teeth carrying its label. Acting on this, we then rebuilt the caries stage to localise lesions rather than label teeth—something the constant cannot do at all—and on the same 337 external radiographs it beat the constant by 9.3 percentage points while flagging fewer teeth than it. Five prior attempts to improve the classifier, leaving the task unchanged, had all failed. We do not conclude that automated caries screening fails. We conclude that a tooth-level score alone cannot establish that a system is reading the radiograph—a pipeline can clear an image-blind floor on the data it was built on and be indistinguishable from it elsewhere—and that four things a constant cannot do—identifying which tooth, reporting a confidence that behaves like a probability, declining when uncertain, and being measured on outside institutions—are where the evidence for a working system has to come from.**

***Index Terms*—panoramic radiography, dental caries, data leakage, evaluation methodology, image-blind baselines, external validation, model calibration, selective prediction, domain shift**

I. Introduction

The clinical setting

A panoramic dental radiograph—an *orthopantomogram*, or OPG—is a single X-ray image that captures the whole upper and lower jaw in one exposure. It is the most common extraoral radiograph taken in dentistry, because one image at a low radiation dose shows every tooth, the bone that supports them, and the jaw joints. Dentists use it to survey a mouth quickly: to find teeth that need treatment, to plan extractions and implants, and to check for disease a patient has not yet noticed.

Two diseases dominate that survey. **Dental caries** is tooth decay: acid produced by bacteria dissolves the hard tissue of the tooth, first the enamel surface and then the softer dentine beneath. On a radiograph it appears as a dark (radiolucent) area where solid tooth should be. Caught early, it is treated with a filling; missed, it reaches the nerve and requires root-canal treatment or extraction. **Periodontal disease** is inflammation of the tissues that hold a tooth in place. In its advanced form it destroys the surrounding bone, and that destruction—**periodontal bone loss**—is what a radiograph can show: the bone level around the tooth root sits lower than it should. It is the leading cause of tooth loss in adults, and it is painless until late.

Both are common, both are visible on an OPG, and both are missed. Panoramic images are geometrically distorted, teeth overlap one another, and a busy clinician reads many of them quickly. Automated screening is attractive for exactly this reason: not to replace the dentist's judgement, but to make a second pass over an image a human has already read, and flag what may have been overlooked.

What automated screening currently reports

Deep learning on OPGs has converged on a standard decomposition: detect the individual teeth, then classify each one. Systems built this way report caries and periodontal findings at accuracies that read as clinically useful [1]–[3]. The result is usually reported as a **tooth-level** score: of the teeth in this

mouth, which ones does the system say are diseased, and how well does that list agree with the expert's.

The evaluation problem, by example

Consider a program that contains no neural network, no image processing, and no pixels—only a fixed list of eight tooth codes: the first and second molars in all four corners of the mouth. It emits that same list for every radiograph it is ever shown. It cannot see.

On our development data, that program achieves an F1 score of **0.320** on the tooth-level caries question. Two general-purpose vision–language models, which do read the image, achieve 0.246 and 0.149. Our own six-stage pipeline, asking of each tooth in turn whether it is carious, achieves **0.362**—four points above a list that cannot see, on a corpus its tooth detector was trained on.

The reason is not that these systems are broken. It is that caries is not spread evenly around the mouth: molars have deep grooves, are hardest to clean, and carry most of the disease. A tooth-level score asks which members of a fixed set of 32 tooth positions are affected—a question that population statistics answer partially correctly without reference to any particular patient. A fixed answer therefore starts with a substantial score already in hand, and the room left above it is small.

Why that matters

This is a statement about *evidence*, not about anyone's model. A published tooth-level score is not wrong. But on its own it cannot distinguish a system that recognises decay in this patient's radiograph from a system that has learned where decay usually is. The two produce similar numbers and very different clinical objects: only the first can find the lesion in the place the population statistics do not expect, which is precisely the lesion a hurried human reader also missed.

The natural response is to check what the floor is. Reading the 49 task-relevant papers of our own survey of this area, we noticed **two** that report validation on data from an institution other than the one that produced their training set, and none that reports an image-blind or otherwise uninformed baseline—an observation from that reading, not a systematic count.

What we did

We measured that floor on our own work rather than anyone else's. We first built a system worth testing: a six-stage pipeline performing tooth detection, tooth numbering, caries classification and a per-tooth abnormality flag, with confidence intervals, calibrated probabilities, a threshold for declining uncertain cases fixed before the data were seen, and evaluation on four hospitals that contributed no training data. We then constructed a ladder of opponents, each given strictly less information than the one above, and scored every rung with one scorer on one set of images.

Testing on outside institutions matters for a specific reason. A model learns the scanner, the exposure settings, the patient population and the annotation habits of the place its training data came from, along with the disease. Only data from somewhere else can separate the two. We report every external result per hospital, because pooling four sites into one number conceals exactly the variation the exercise exists to expose.

Contributions

1) An **image-blind constant-baseline protocol** for tooth-level OPG evaluation, fitted without leakage and applied across two unrelated metric families and three cohorts, one of them four hospitals.
2) A **vision–language-model baseline with the controls that make it interpretable**, including what these models answer when shown an image containing no anatomy at all.
3) **External validation on four hospitals and a second public benchmark**, reported per site, with the spread that pooling conceals.
4) **Calibration and selective-prediction** results, each with a control that isolates the mechanism responsible, and each reported with its cost rather than only its gain.
5) An honest characterisation of the pipeline, including a fourth stage whose label is not the one it was named for and whose first score did not survive a split by radiograph, and a tooth detector whose numbering result is strong only on the corpus it was trained on and which was adopted over a pre-registered guard it failed.

We are explicit about what we do not claim. We do not assert that published caries results are incorrect, and we do not claim our own system is better than any of them. We argue something narrower: that a tooth-level number reported without a floor cannot be distinguished from an anatomical prior.

II. Related Work

A. Detection and classification on panoramic radiographs

Detector choice for this data is well studied. A head-to-head evaluation on dental panoramics reports YOLOv11 outperforming its successor generation for multi-class dental detection, and observes an asymmetry that matters here: performance is strong on well-defined objects such as crowns and implants and markedly weaker on subtle radiolucent pathology, including caries and bone loss [4]. YOLO-family detectors are widely applied to OPGs more generally [5]–[7]. Caries and periodontal classification have been approached with dual-network frameworks [1], dedicated convolutional classifiers [3], and contrastive feature-pyramid designs [2], alongside broader detection and segmentation work [8], [9].

B. A published benchmark, and what a stock detector reaches on it

One of our corpora, CariesXrays [2], ships its own detection baselines, which lets us place a number of ours beside prior work on the same data rather than only against our own arms. We trained a stock YOLOv11l on that corpus's training portion, changing nothing for the task, and report COCO AP against the authors' table (Table I).

TABLE I
LESION-DETECTION AP (%) ON CARIESXRAYS. ROWS 1–4 ARE REPRODUCED FROM TABLE 1 OF [2]; ROW 5 IS OURS. **THE COMPARISON IS INDICATIVE, NOT HEAD-TO-HEAD**—SEE THE CAVEAT IN THE TEXT. ROWS 1–3 ARE THE THREE STRONGEST OF THE TEN BASELINES THE AUTHORS REPORT.

Method	Backbone	AP	AP_{50}	AP_{75}
Faster R-CNN	ResNet-50	39.9	78.0	37.8
YOLOx	CSPDarkNet	40.5	81.3	36.1
Conditional-DETR	Transformer	42.2	80.6	40.4
FPCL (theirs)	ResNet-50	**48.2**	84.1	**50.6**
YOLOv11l (ours)	CSPDarkNet	**48.2**	**84.5**	50.0

We do not claim to have matched or beaten FPCL. The authors' 4,800 / 1,200 split is not public—the released repository ships a 0.5 % sample and the data download carries no image sets—so we drew our own seed-fixed split at the same ratio. The two rows are therefore scored on different test images and are not directly comparable. What the comparison supports is weaker and about the benchmark rather than about us: a generic one-stage detector with no caries-specific design lands at the same AP as a task-specific contrastive method and about six points above the strongest of the ten baselines the authors themselves report, so reaching this benchmark's published ceiling does not appear to require a specialised architecture.

Two properties of this row matter later. It is a *lesion-level* detection score, a different quantity from the tooth-level agreement everything else in this paper is measured on. And the checkpoint behind it is the same one that appears in Section VI-C as the single-corpus arm—the arm that reaches 0.694 on this corpus and then falls to 0.086 on another. A strong benchmark number and a failure to transfer are, here, the same model.

C. Tooth numbering, and two different tasks with one name

Tooth numbering is typically reported at very high accuracy—one recent system reports 0.9988 [10], and multi-stage segmentation-and-numbering pipelines report comparable figures [11]. Such numbers are frequently obtained on *pre-cropped* teeth, where the task is a 32-way classification of an image already known to contain exactly one tooth, correctly framed. That is a materially easier problem than numbering under detector output, where teeth may be missed, duplicated or clipped. We report both framings and find the gap between them large.

D. Vision–language models in dentistry

A **vision–language model** (VLM) is a general-purpose model that takes an image and a written instruction and replies in text. Recent work applies general-purpose and domain-adapted VLMs to dental imaging [12]–[15]. They are attractive because they need no task-specific training, and difficult to evaluate for the same reason: a fluent, confident, correctly formatted answer is not evidence that the model perceived anything. We adopt a control from the shortcut-learning literature [16]—present an input containing none of the signal the model purports to read, and observe whether its behaviour changes.

E. External validation and the missing floor

External validation appears uncommon in this subfield: reading the 49 task-relevant papers of our survey, we noticed two that evaluate on an outside institution. Where it is done, degradation is often severe: one dental object-recognition system falls from 95.06 mAP on its own data to 61.83 on a public benchmark [17]. Independent work on periodontal bone loss, evaluated against five clinicians, reports weighted F1 of 0.29 with severe-class F1 of zero [18]—a reminder that in-domain figures can be a poor guide to behaviour elsewhere. To our knowledge no work in this area reports an image-blind baseline, which is the gap this paper addresses.

III. SYSTEM

The finding in this paper is only interesting if the system it is measured against is a serious one. We therefore describe the pipeline before the evaluation that tests it.

A. Architecture

The pipeline maps an OPG to a structured per-tooth record in six stages (Table II).

Stage 1 detects individual teeth with a YOLOv11l detector [19] trained on three corpora (Section IV), emitting one box per tooth.

Stage 2 assigns each detected tooth its clinical name. We use **FDI two-digit notation**, the international standard: the first digit is the quadrant of the mouth (1 = upper right, 2 = upper left, 3 = lower left, 4 = lower right) and the second is the position counting outwards from the midline (1 = central incisor, through to 8 = third molar, the wisdom tooth). Tooth 16 is therefore the upper-right first molar and tooth 47 the lower-right second molar. Numbering matters because everything downstream is reported against it: a caries verdict on the wrong tooth code is a wrong clinical finding, however good the classifier. A ResNet18 [20] head consumes the tooth crop together with four positional features, and the resulting scores are resolved to a one-to-one assignment across the mouth by the Hungarian algorithm [21], so no two teeth receive the same code.

Stage 3 reports caries per tooth. **Stage 4** flags teeth near an expert-marked abnormality (Section VI-F): it was built for periodontal bone loss, but its training label is whether a tooth crop overlaps an expert's abnormality mask on Tufts, and we name it for that label. A second model, meant to grade the flagged teeth, was trained on bands of crop contrast instead; it ships withheld by default, and a retrain on the bands it was named for did not beat an image-blind tooth-position prior (Section VI-F). Stage 3 was built as the same kind of cascade—a binary gate on a tooth crop, then a severity head on the teeth it flags—and is reported that way throughout Section VI. It now defaults instead to a whole-image lesion detector, whose predicted lesion boxes are assigned to teeth before the same severity head runs on the teeth so flagged; the

TABLE II

PIPELINE STAGES AND MODEL SIZES. PARAMETER COUNTS ARE LEARNABLE PARAMETERS READ FROM THE SAVED CHECKPOINTS, NOT FROM THE ARCHITECTURE NOMINAL. STAGE 3A HAS TWO INTERCHANGEABLE IMPLEMENTATIONS; THE WHOLE-IMAGE LESION DETECTOR IS THE DEFAULT AND THE PER-CROP GATE IS RETAINED AS A VARIANT (SECTION III-B).

Stage	Model	Parameters
1 Tooth detection	YOLOv11l	25.31 M
2 FDI numbering	ResNet18 + 4 pos. feats	11.19 M
3a Caries localisation	YOLOv11l detector	25.31 M
variant	*ResNet18 per-crop gate*	*11.18 M*
3b Caries severity	ResNet18	11.18 M
4a Abnormality flag	ResNet18	11.18 M
4b Severity (withheld)	ResNet18	11.18 M
Total		**95.34 M**
Caries path (stages 1–3)		*72.99 M*

Running the Stage-3a gate variant in place of the detector gives 81.21 M total and 58.85 M for the caries path.

per-crop gate is retained, unchanged, as a selectable variant. Section III-B states why the formulation changed and what that costs.

Stages 5 and 6 fuse the per-stage outputs on the FDI key and emit the per-tooth record.

The complete system runs at **1.70 s per image on a laptop CPU** with no GPU present, measured with the Stage-4 severity grade computed; the report now withholds that grade by default, so the figure is conservative. Both figures become relevant in Section VI.

B. Stage 3a: two formulations of the same clinical question

The two Stage-3a implementations are not two models for one task. They pose two different tasks, and the distance between them is the subject of Section VI.

The **per-crop gate** asks, of each detected tooth in isolation, whether that tooth is carious. Posed that way, the question is dominated by its base rate. Caries is uncommon per tooth and concentrated in the molars, so a rule that names the molars and never looks at the image is a serious competitor—serious enough, on our data, to tie the trained gate pooled over four external hospitals and to beat it outright on a second external benchmark.

The **lesion detector** asks instead where the lesions are in the radiograph. A YOLOv11l detector runs once over the whole image and predicts caries lesion boxes directly; each predicted box is assigned to the detected tooth box it overlaps most, by argmax containment, and those teeth are the positives. The severity head is unchanged and runs on the teeth so flagged. A model that must place a lesion cannot score well by reproducing a prior over tooth positions, because naming the molars produces no boxes. The data bear that out: on the 164 DENTEX images it flags 624 tooth codes, 43.9 % of them carious and 71.6 % molars, against the constant's 1,312, 21.8 % and 100 %.

Training pools two corpora: 4,800 CariesXrays images with their lesion boxes, and 650 DENTEX images whose caries annotations are coarser, whole-tooth boxes—a different DENTEX subset from the `disease/` images scored in Section VI, which carry per-lesion polygons—the latter replicated to equal weight (9,600 samples per epoch, seed 42, zero cross-corpus byte overlap verified). Detection confidence is fixed at 0.25 *a priori* and was never tuned against any cohort. A CariesXrays-only arm was trained and scored alongside the pooled one; Section VI reports both, because the difference between them is the result.

What the reformulation costs, stated plainly. Both training corpora were external test data for the gate and are development data for the detector. Of the cohorts in Section IV, only the Chinese one is external to both, so the detector is supported by one external cohort where the gate had two. We regard that as the honest price of the change rather than a detail: it is why the detector's external result is reported on its own below, and not averaged with the rest. The decision rule, the comparators, the confidence and the acceptance bar were written down and committed before any arm was trained, and are reported where they fell.

C. The fourth stage, described by its label

Stage 4 is a full component of the implemented system, and we keep it in the paper described by what it was trained on rather than by what it was built for. Its first score is the clearest demonstration available to us that an in-domain figure measured on a leaky split is not evidence of a usable model. It is reported throughout as a *secondary* analysis (Section VI-F). The primary evaluation—the baseline ladder—concerns caries only, because caries is the task for which we hold annotations on all three cohorts.

IV. DATA

A. Development corpora

The **Tufts Dental Database** [22] (1,000 OPGs) is held under a frozen 700/150/150 image-level split for tooth detection, which the retrained Stage-4 flag shares; the first Stage-4 model was trained on tooth crops under a crop-level split instead (Section VI-F). The caries branch and the primary evaluation use **DENTEX** [23]. Tooth detection trains on three corpora sampled in equal proportion: the Tufts training split, DENTEX's enumeration subset, and a third public corpus. FDI numbering trains on the first two, with the third used only for an ablation (Section VI-D). Because the detector has seen DENTEX, every tooth-level result on the 164 scored DENTEX images is an *in-domain* result for the front end, held out by image but not by corpus; we label it so wherever it appears.

All evaluation splits are **image-level**; the first Stage-4 model's crop-level split is the one exception, and Section VI-F reports what it cost. Two crops from one radiograph are not independent observations, and a tooth-level split would place the same mouth in both training and test data.

Two labelling conventions were established by measurement rather than assumed, because a silent error in either would have trained a plausible but wrong model. Tufts numbering is left–right **mirrored** relative to Universal notation (Tufts "1"

TABLE III

EVERY CORPUS USED, AND WHAT IT WAS USED FOR. "EXTERNAL" MEANS NO IMAGE FROM THAT CORPUS CONTRIBUTED TO TRAINING OR TUNING ANY MODEL IN THE PIPELINE. EVALUATION SPLITS ARE IMAGE-LEVEL THROUGHOUT, EXCEPT THE FIRST STAGE-4 MODEL'S (SECTION VI-F).

Corpus	Role	Images	Annotation	Split used	External?	Fits baseline?
Tufts [22]	Detection, Stage 4	1,000	tooth boxes; expert masks	700/150/150 frozen	no	no
DENTEX [23]	Caries; numbering	250 [a]	per-lesion polygons	164 scored / 86 excluded	no	78 of the 86
Kaggle dual-labelled	Detection; ablation	499	per-tooth boxes, FDI	18,372/1,961/1,994 crops	no	no
Ningxia [24]	External test, locked	337 [b]	raster lesion masks	test1–test4, all scored	**yes**	no
CariesXrays [2]	External test [c]	6,000	per-lesion boxes	1,200 scored	**yes** [c]	no

[a] The DENTEX `disease/` subset, which is the portion carrying per-lesion annotation. The tooth detector and the numbering head train on DENTEX's separate enumeration subset (634 images), never on a scored image. [b] Caries-annotated subset. [c] 13,784 annotated caries instances counted in the release. External to every component except the Stage-3a lesion detector, which trains on the other 4,800 images and chose its checkpoint on the 1,200 scored here, its validation split (Section III-B); those carry 2,815 lesions.

corresponds to FDI 28, not 18), confirmed three independent ways. The third corpus was checked the same way, using the two-corpus numbering head as an independent arbiter over 915 teeth: agreement was **80.3 %** reading its labels as written, against **1.7 %** reading them mirrored. Its labels are therefore standard FDI and are used as written.

B. Leakage control, and how it was established

The DENTEX `disease/` subset contains 250 annotated images. Our own models had already trained on some of them, so those must never be scored. Overlap was determined by **MD5 hashing every image and matching byte-identical files**, not by filename and not by assertion: 85 images are byte-identical to an image used in numbering-head training and 1 to an image used in detector training, giving **86 excluded** and **164 scored**. The record is stored as a manifest and re-checked at run time.

The image-blind constant is fitted on those excluded images and applied to the scored ones. Of the 86, **78 carry at least one caries annotation**, and those 78 are the fitting set. The two sets are disjoint by construction, and the fitting routine asserts the empty intersection before it runs.

We note the residual limitation honestly: DENTEX ships no patient identifiers, so we can establish image-level and byte-level separation but not patient-level separation. If one patient contributed two radiographs under different filenames, we could not detect it.

C. External data

Four hospitals from an independent Chinese cohort [24] (denoted test1–test4) serve as locked external test data: never trained on and never tuned against. They have been scored in at least 25 recorded runs across tooth-detector and caries-stage variants, and three of their uses go beyond testing: the current tooth detector was chosen with its result there in view; that result was a veto in the detector's pre-registered adoption rule; and the cohort was one of the three that decided the lesion detector's adoption (Section VI-C). It is therefore the only cohort the lesion detector neither trained on nor chose its checkpoint on, but not one untouched by its selection. For caries this is 337 images carrying 1,013 annotated lesions.

One property of that cohort must be stated because it invites an obvious question. The cohort ships its own train/test division, and **test2 is drawn from the same centre as that cohort's own training portion**, while test1, test3 and test4 are cross-centre. *We never trained on any Ningxia data*, so all four splits are equally external to our system; the label describes the source cohort's design, not our exposure. We report it because test2 is also the site where our pipeline scores highest, and a reader is entitled to know that the two facts coincide.

A second external cohort is the published **CariesXrays** benchmark [2]: 6,000 hospital-scale panoramic radiographs from 5,380 patients, annotated by dental professionals and released with its own detection baselines (Section II-B). We count **13,784** caries instances in the release we obtained; the paper states 13,778 and the authors' repository 13,783. We report our own count and note the discrepancy rather than adopt either figure, since nothing here depends on which is right. We score the 1,200 images that serve as the Stage-3a lesion detector's validation split—no gradient came from them, but its training stopped and chose its checkpoint on them, so they are external for the gate and the constant and not for the detector—carrying **2,815 lesions**—more than the development and first external cohorts combined—and verified them leakage-free against every corpus above, both byte-wise and perceptually. Its licence is not stated by its authors; it is released publicly alongside the work cited here and used under that citation. One property bounds what it can support and is reported rather than omitted: the corpus does not document whether its annotation is *exhaustive*, so a flagged tooth carrying no annotated lesion is not a confirmed false positive.

D. Crop geometry and pre-registration

Classifier inputs are expanded crops resized to 224×224. Expansion is fixed per branch and identical across corpora: symmetric $1.5\times$ for numbering, and $1.5\times$ width by $2.0\times$ height for Stage 4, since teeth are roughly three times taller than wide and horizontal over-expansion bleeds into neighbouring teeth.

Operating points are fixed *a priori* and reported where they fall rather than tuned against evaluation data. One consequence concerns the tooth detector, does not flatter it, and is reported here rather than omitted. The detector was trained on three corpora to repair tooth numbering outside Tufts, and the rule

for adopting it was registered before it was trained: end-to-end numbering had to improve on DENTEX, external caries agreement had not to fall, and box mAP@0.5 on the frozen Tufts test split had to stay within four points of a Tufts-only detector trained by the same recipe. The first two conditions were met. The third was **missed**: **0.587** against the control's 0.723, a loss of 13.6 points on the corpus the other two had been added to. The bar was not moved after the fact, and an earlier two-corpus attempt had failed the same kind of guard and been set aside. The detector was adopted regardless, by an explicit, dated decision taken with that result in view, for the numbering and caries behaviour reported in Section VI. We state this so that those figures are read with their cost beside them, and because a paper that asks others to report what does not favour them should not conceal a rule its authors overrode.

That frozen Tufts test split carries one documented flaw: one test image is byte-identical to one training image. The mAP figures above include it, for both detectors alike; the Tufts numbering figures in Section VI exclude it.

V. Evaluation Protocol

A. The ladder

Four systems answer one question—*which teeth in this radiograph show caries?*—on the same 164 leakage-free DENTEX images, scored by one scorer per metric family. Each rung receives strictly less information than the one above.

1) **Pipeline**: the six stages of Section III, scored once with each Stage-3a implementation.
2) **VLM, large**: Qwen2.5-VL-32B-Instruct [25], zero-shot; 33.45 B parameters as loaded.
3) **VLM, small**: Qwen2.5-VL-7B-Instruct, zero-shot; 8.29 B parameters as loaded.
4) **Constant**: a fixed set of FDI codes emitted for every image. **It reads no pixel of the radiograph.**

B. How the constant is built

The constant is not hand-picked. We ran a greedy search over tooth codes on the 78 held-out fitting images, adding whichever code most improved F1 until no addition helped, and then froze the result. It selected $\{16, 17, 26, 27, 36, 37, 46, 47\}$—the first and second molars of all four quadrants.

Two properties are worth stating. First, the search never sees a scored image. Second, the frozen answer transfers *downwards*: it scores F1 0.392 on the images it was fitted on and 0.320 on the images it is scored on, which is what an honestly held-out fit should do.

We additionally report a variant fitted directly on the scored set. It is labelled an *oracle* throughout, is an upper bound on image-blind performance rather than a deployable baseline, and is not claimable.

C. Fairness

A rigged baseline is worse than no baseline. The VLMs are given the FDI numbering scheme in the prompt, told explicitly that teeth may be absent, instructed to answer with codes alone, and parsed permissively, so that neither notation nor formatting is scored as clinical error. Few-shot exemplars, where used, are drawn strictly from the leakage-excluded images—which our own pipeline trained on—so the VLM receives free supervision our pipeline does not. Every affordance favours the opponent.

D. Metrics

We use two metric families, chosen because they fail differently. Each is defined in words before symbols, and each is reported with its limitation.

FDI-set F1 treats each image as a set-prediction problem: *of the tooth codes the expert marked carious, how many did the system name, and how many of the codes it named were not marked?* Precision is the fraction of predicted codes that are correct, recall the fraction of annotated codes that were found, and F1 their harmonic mean; all three are micro-averaged over teeth pooled across images. Its limitation is the subject of this paper: set membership over a fixed 32-code vocabulary can be answered partially from population statistics alone.

Lesion-localisation agreement asks, of every expert-annotated lesion, *was a tooth flagged that overlaps it in space?* It requires spatial correspondence, which set membership does not. On DENTEX, a lesion polygon is matched to a predicted-positive tooth box by greedy intersection-over-union at a threshold of 0.30 with a centroid-containment fallback. On the external cohort the matcher is centroid containment alone: those lesions are small focal blobs (median area 731 px) inside much larger tooth boxes ($\sim$25,000 px), and every true match scores an IoU of 0.015–0.124, below any usable threshold. **The two matchers are not interchangeable, and internal and external agreement figures are therefore not directly comparable to one another**; each is comparable across systems within its own cohort, which is what the comparison requires.

One property of the constant under this metric must be stated, because it shapes every localisation table below. The constant chooses nothing from the pixels, but its eight codes have to be put somewhere on the image, and they are put there by the pipeline's own tooth boxes and numbering. The constant therefore inherits the pipeline's front end: when that front end finds and names more teeth, the constant's agreement rises with the pipeline's. A change in an agreement figure is evidence about the caries stage only to the extent that it exceeds the constant's change on the same images. On Tufts we measured where the eight codes land: 92.0 % $[89.58, 94.06]$ of the constant's codes sit on a true first or second molar, and they reach 81.7 % $[77.72, 85.35]$ of the true ones. Under this metric it is therefore a constant placed by the pipeline's front end; "image-blind" describes only the F1 constant, which is scored on tooth codes alone.

Lesion agreement has no negative class by construction—a deliberate property, since the external cohort contains no healthy controls—so it is a recall-shaped measure, and flagging more teeth can only raise it. We therefore always report it

alongside the number of teeth flagged and the predicted-positive match rate, and we include an upper-bound arm that flags *every* detected tooth so the reader can see what saturating the metric costs.

Because these two families count different things, their denominators differ on the same images: the 164 scored radiographs carry **479 annotated caries lesion polygons** distributed over **477 distinct teeth**. The difference is exactly two teeth that each carry two separate annotated lesions. Both numbers are correct; they are different units, and each table states which it uses.

E. Controls

For the VLMs, we present a **blank grey image** and a **random-noise image** under the identical prompt. Neither contains a tooth. For selective prediction, we **shuffle the confidences**, breaking the link between confidence and correctness while preserving both marginal distributions; a gain that survives shuffling comes from discarding items rather than from ranking them. For the localisation metric, the flag-every-tooth arm bounds the achievable range.

F. Statistical analysis

All confidence intervals are 95 % percentile intervals from a **cluster bootstrap resampled over images**, 10,000 resamples, seed 42. Differences between two systems evaluated on the same images are computed as paired differences within each resample.

Images, not teeth, are the resampling unit, and this is the choice to defend. The 936 teeth in the numbering evaluation come from 164 radiographs. Teeth within one image share a patient, a scanner, an exposure and an operator, and errors are correlated within an image—a mis-numbered quadrant mis-numbers several teeth at once. Treating 936 correlated teeth as 936 independent observations would produce intervals far too narrow. Where a metric's clustering differs, the number of clusters is reported with it.

One group of quantities carries no interval and we say so rather than implying otherwise: detection mAP and the in-domain scores of the two classifier branches, whose per-item prediction dumps were not saved and which would need a re-evaluation pass to bootstrap.

VI. Results

A. The ladder on development data

Fig. 1 summarises the comparison; Table IV gives the set-prediction result exactly.

Read from the bottom, the ordering is inverted with respect to the information each system receives. Both vision–language models see the radiograph and score below a list that does not, by $+7.32$ percentage points, 95 % CI $[+3.42, +11.23]$, and by $+17.04$ $[+12.66, +21.22]$. The gap between the frozen constant and the test-set-fitted oracle is one percentage point—so almost all of the image-blind score is available without any access to the evaluation data.

TABLE IV

Tooth-level caries prediction. **Dataset:** 164 leakage-free DENTEX images, in-domain for the tooth detector. **Task:** name the carious teeth. **Metric:** FDI-set precision/recall/F1, micro-averaged. **Unit:** tooth code; $n = 477$ annotated carious teeth. One scorer for all arms. The pipeline is scored with each Stage-3a implementation; its parameter counts are for the caries path.

System	Precision	Recall	F1	Params
Pipeline, lesion detector	0.439	0.574	**0.498**	72.99 M
Pipeline, per-crop gate	0.237	0.765	0.362	58.85 M
Constant (never reads image)	0.218	0.600	0.320	0
Qwen2.5-VL-32B-Instruct, 0-shot	0.250	0.243	0.246	33.45 B
Qwen2.5-VL-7B-Instruct, 0-shot	0.248	0.107	0.149	8.29 B
Oracle constant (test-fitted)	*0.245*	*0.505*	*0.330*	*0*

TABLE V

Lesion localisation. **Dataset:** the same 164 DENTEX images. **Task:** flag a tooth overlapping each annotated lesion. **Metric:** agreement (recall-shaped; no negative class). **Unit:** lesion; $n = 479$ annotated caries lesions. Teeth flagged is reported because agreement alone rewards flagging more.

System	Agreement	Teeth flagged	Match rate
Pipeline, per-crop gate	**0.802**	1,439	0.267
Constant	0.630	988	0.306
Pipeline, lesion detector	0.595	603	0.473
All teeth (ceiling)	*0.998*	*4,132*	*0.116*

Our pipeline scores above the constant, and how far above depends on the question its caries stage asks. With the per-crop gate it leads by $+4.27$ points, CI $[+1.58, +7.05]$: it recovers more of the carious teeth than the constant (recall 0.765 against 0.600) at much the same precision, by naming 1,537 teeth where the constant names 1,312. With the lesion detector it leads by $+17.80$ points $[+13.48, +22.02]$, and does so on precision—0.439 against 0.218—while naming fewer than half as many teeth as the constant. Section VI-C returns to that difference. Both leads are measured on a corpus the tooth detector was trained on (Section IV), and the question the remaining tables ask is what is left of them once that advantage is removed.

A single metric is a single view. We therefore repeated the comparison under lesion localisation, which demands spatial correspondence (Table V).

Here the per-crop pipeline leads the constant by **17.12 percentage points**, 95 % CI $[+12.15, +21.89]$. Part of that lead is volume: it flags 46 % more teeth than the constant, at a lower match rate (0.267 against 0.306), and this metric rewards flagging—the arm that flags every detected tooth reaches 0.998. The lesion detector flags fewer teeth than either, at the highest match rate of the three (0.473), and its agreement is not separable from the constant's: -3.55 points, CI $[-10.11, +2.64]$.

On development data, then, the pipeline is above the image-blind constant under one metric, and above or level with it under the other depending on its caries stage. Neither statement is yet evidence of perception, because the front end that places

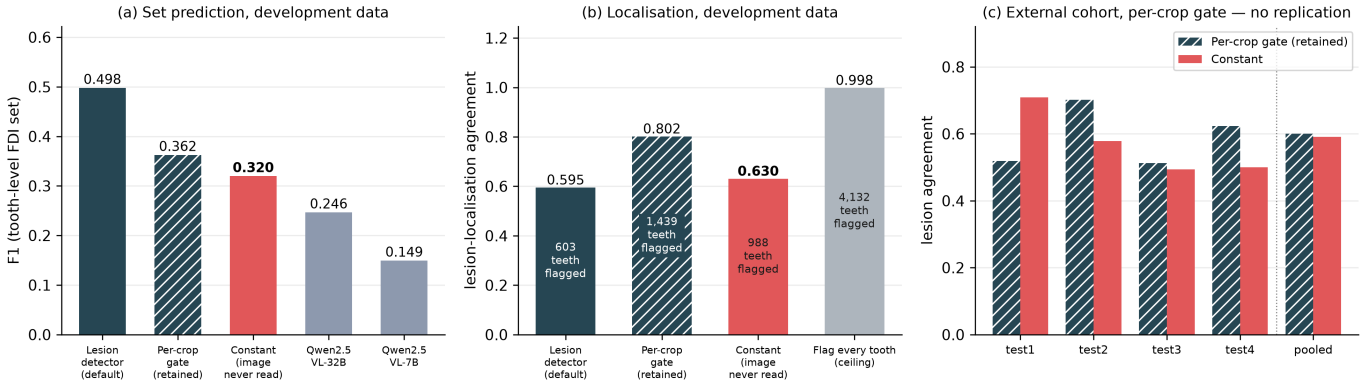


Fig. 1. The constant-baseline comparison. **(a)** Tooth-level FDI-set F1 on 164 leakage-free DENTEX development images (477 annotated carious teeth). Both vision–language models score below a constant that never reads a radiograph; the pipeline scores above it, by an amount that depends on its caries stage. **(b)** A second, unrelated metric on the same images (479 annotated lesions), requiring the flagged tooth to overlap the lesion in space. Teeth flagged is annotated inside each bar because agreement is recall-shaped, and the flag-everything arm shows the metric's upper bound. **(c)** The external cohort (337 images, 1,013 lesions, four hospitals), where the development lead does not replicate: the difference swings 31 points between sites and vanishes when pooled. Panels (a) and (b) are in-domain for the tooth detector and are not comparable to (c): different cohorts and different lesion matchers (Section V). Every value is read from the saved evaluation artefacts at render time rather than transcribed.

TABLE VI

EXTERNAL VALIDATION. **DATASET:** FOUR HOSPITALS, 337 IMAGES. **TASK:** FLAG A TOOTH CONTAINING EACH LESION CENTROID. **METRIC:** LESION AGREEMENT BY CENTROID CONTAINMENT—A DIFFERENT MATCHER FROM TABLE V, FOR THE REASON GIVEN IN SECTION V. **UNIT:** LESION; $n = 1{,}013$. Δ IS CONSTANT MINUS PIPELINE.

Hospital	Centre	Lesions	Pipeline	Constant	Δ (pp)
test1	cross	316	0.519	**0.709**	+19.0
test2	same [a]	349	**0.702**	0.579	−12.3
test3	cross	162	0.512	0.494	−1.9
test4	cross	186	0.624	0.500	−12.4
Pooled	—	1,013	0.600	0.591	−0.89

[a] Same centre as the *source cohort's own* training portion. No Ningxia data was used to train or tune our system; all four splits are equally external to us (Section IV).

every flagged tooth—the constant's included—was trained on this corpus.

B. The ladder on external data

We then ran the same comparison on four hospitals the system had never seen (Table VI). The rows score the pipeline with its per-crop gate, the stage this comparison was designed around; Section VI-C gives the lesion detector on the same images.

The development-set lead does not replicate. Pooled, the difference is -0.89 percentage points with 95 % CI $[-5.57, +3.86]$: the two systems are statistically indistinguishable, with the pipeline nominally ahead by an amount the interval cannot separate from zero.

The per-site spread is the more useful observation. The difference swings from $+19.0$ points at test1 to -12.3 at test2, both intervals excluding zero—a 31-point range across hospitals of a single cohort—and the pooled figure conceals this entirely. test1 is the constant's best site by a wide margin and one of the pipeline's two worst. We do not have an explanation for that, and we report it unexplained rather than omit it.

TABLE VII

THE SAME COMPARISON ACROSS THREE COHORTS: THE PIPELINE WITH ITS PER-CROP GATE AGAINST THE CONSTANT. **METRIC:** LESION AGREEMENT BY CENTROID CONTAINMENT FOR THE TWO EXTERNAL ROWS; THE DEVELOPMENT ROW USES THE MATCHER OF TABLE V AND IS NOT COMPARABLE IN ABSOLUTE VALUE. **UNIT:** LESION. Δ IS CONSTANT MINUS PIPELINE; INTERVALS ARE PAIRED CLUSTER BOOTSTRAPS OVER IMAGES.

Cohort	Lesions	Pipeline	Constant	Δ (pp), 95 % CI
DENTEX (dev.)	479	**0.802**	0.630	-17.12 $[-21.89, -12.15]$
Ningxia (4 hosp.)	1,013	0.600	0.591	-0.89 $[-5.57, +3.86]$
CariesXrays	2,815	0.379	**0.604**	$+$**22.42** $[+$**20.18**$, +$**24.62**$]$

A second external cohort makes the point sharper still. We repeated the identical comparison, with the same matcher and the same frozen system, on the published CariesXrays benchmark [2]—1,200 images carrying 2,815 lesions, more than the development and Ningxia cohorts combined (Table VII). There the constant does not merely tie: it wins by **22.42 percentage points**, 95 % CI $[+20.18, +24.62]$.

Two observations follow, and the second is the one that matters. First, **volume explains part of one result and none of the other**. On the development set the pipeline's lead came with 46 % more flagged teeth at a lower match rate. On CariesXrays the roles reverse and the explanation fails: the constant flags 46 % more teeth *and* attains the higher hit rate (0.200 against 0.188). It is ahead on both axes.

Second, **the size of the gap is a property of the cohort, not of the systems**. The constant's own agreement on CariesXrays, 0.604, is the share of that cohort's lesions lying on the eight teeth it names. Where caries concentrates in the molars to that degree, an anatomical prior is close to optimal by construction.

Taken together the three cohorts place the same pipeline 17 points above the constant, level with it, and 22 points below it, depending only on which cohort is scored—and the

TABLE VIII

STAGE 3A REFORMULATED AS LESION DETECTION, UNDER THE CURRENT TOOTH DETECTOR. **METRIC:** TOOTH-LEVEL LESION AGREEMENT, MATCHED EXACTLY AS IN TABLE VII. **UNIT:** LESION. WITHIN A COHORT ALL THREE ARMS SCORE THE SAME IMAGES, AND EACH ROW IS THE SAME COHORT AS IN TABLE VII; INTERVALS ARE PAIRED CLUSTER BOOTSTRAPS OVER IMAGES (10,000 RESAMPLES). LESION COUNTS ARE 479, 2,815 AND 1,013 BY ROW.

Cohort	Gate	Const.	Lesion	Δ vs const. (pp)
DENTEX[d]	0.802	0.630	0.595	-3.55 $[-10.11, +2.64]$
CariesXrays[d]	0.379	0.604	**0.757**	$+$**15.38** $[+$**12.96**$, +$**17.73**$]$
Ningxia[e]	0.600	0.591	**0.684**	$+$**9.28** $[+$**4.72**$, +$**13.73**$]$

[d] Development data for the lesion detector: a corpus it trained on, scored on images it did not train on; for CariesXrays those are its validation split, on which its checkpoint was chosen. [e] External: never trained on or tuned against, but one of the three cohorts that decided this detector's adoption.

one cohort on which it leads is the one its tooth detector was trained on. The swing across a single cohort's four sites is 31 points; the swing across cohorts is 40. **A tooth-level agreement figure reported on one cohort therefore carries almost no information about a system's perceptual ability**, and a favourable one on development data carries least of all.

C. Changing the task, not the architecture

The external tables above score the per-crop gate of Section III-B. Five attempts to repair that gate directly—contrast equalisation, crop expansion, position-balanced sampling, a retrained gate, and threshold tuning—all returned intervals spanning zero. We therefore changed the question rather than the model, and replaced the gate with the whole-image lesion detector. The decision rule was registered and committed before any arm was trained: to be adopted, the detector had to beat both the constant and the gate, on at least two of three cohorts, at no more than $1.5\times$ their rate of flagging teeth.

The rule was met on two cohorts of three. It was applied to the pipeline as it then stood, with a tooth detector trained on Tufts alone: the lesion detector led the constant by $+12.2$ points $[+10.1, +14.3]$ on CariesXrays and by $+10.2$ $[+6.3, +14.0]$ on Ningxia, and tied it on DENTEX at -3.3 $[-8.4, +1.3]$. The bar was not moved after the numbers were seen. Table VIII reports the same three arms under the current tooth detector. It is a description, not a second verdict, and the pattern is the one the rule was met on: ahead on both external cohorts, level on DENTEX. We report DENTEX as a tie everywhere it appears.

Four things about this result are worth more than the margin itself.

It is not a volume effect. The objection we raised against the constant applies here in reverse, so we apply it. The detector flags *fewer* teeth than either comparator on every cohort—2.16 per image against the constant's 7.05 on CariesXrays, 4.35 against 7.15 on Ningxia—so it is not buying agreement by flagging more.

It holds, pooled, on the one cohort that is genuinely external. The Ningxia row of Table VIII scores the same 337 images, the same 1,013 lesions and the same constant as the Ningxia row of Table VII. On that fixed comparison the gate was 0.89 points ahead of the constant and could not be separated from it; the detector is 9.28 points ahead with an interval excluding zero. Per hospital it is separated from the constant at test2, test3 and test4 ($+16.91$ $[+9.17, +24.62]$, $+11.73$ $[+1.07, +21.34]$ and $+13.98$ $[+2.34, +24.34]$) and not at test1 (-3.16 $[-10.56, +3.97]$), the hospital where the constant is strongest. It was the first arm in this work to beat the constant on data it had never seen. Because Ningxia was one of the three cohorts that decided the detector's adoption, this lead is conditioned on that choice: its sign is robust, its size optimistic. Under the previous tooth detector the same two comparisons were $+0.79$ and $+10.07$ points; only on DENTEX did a sign change with the detector.

The training mixture, not the detector, is what transfers. A second arm trained on CariesXrays alone was scored beside the pooled one when the rule was applied. It reached 0.694 $[0.674, 0.714]$ on CariesXrays, a shade above the pooled arm's 0.684 on that pipeline, and then collapsed to 0.086 $[0.055, 0.119]$ on DENTEX and 0.240 $[0.209, 0.272]$ on Ningxia—far below even the constant. Judged on its own corpus the two arms look like the same model, and the single-corpus one looks slightly better. Judged anywhere else, it does not work. We could not have told them apart without the external cohort, which is the argument of this paper applied to our own result.

One comparison went against it when the tooth detector changed, and we print it for that reason. On DENTEX lesion agreement the per-crop gate now scores above the lesion detector, 0.802 against 0.595 (-20.67 points, CI $[-26.39, -15.06]$), where the two had been level. The gate reaches that figure by flagging 1,439 teeth to the detector's 603; on FDI-set F1, which charges for false positives, the order is reversed (0.362 against 0.498). Which of the two a reader should prefer depends on what a wrongly flagged tooth costs, and we do not settle that here.

D. What the constant cannot do

A constant answers a set-membership question. It cannot perform tasks whose output is not a fixed list, and the pipeline's results on those tasks are correspondingly meaningful.

Tooth numbering. End-to-end FDI-exact accuracy on DENTEX is $92.8\,\%$ ($869/936$ teeth), $95\,\%$ CI $[90.6, 94.9]$. It is an **in-domain** figure and we report it as one: the 164 images are held out, but the tooth detector was trained on other images of the same corpus. Four of the 936 teeth ($0.4\,\%$) are never detected, and of the 932 that Stage 1 does find, Stage 2 numbers **93.2 %** correctly. It is this second figure that compares to the pre-cropped accuracies in Section II.

On the held-out Tufts test split—3,760 teeth, every tooth in the mouth rather than only those carrying a finding—the same pipeline reaches **63.5 %** end to end, CI $[59.3, 67.5]$: $89.3\,\%$ of teeth are detected and $71.1\,\%$ of those numbered correctly. Tufts too is a corpus the detector was trained on. On UFBA-425, which neither the detector nor the numbering head trained on, the pipeline reaches $44.5\,\%$ end to end, CI $[42.39, 46.62]$

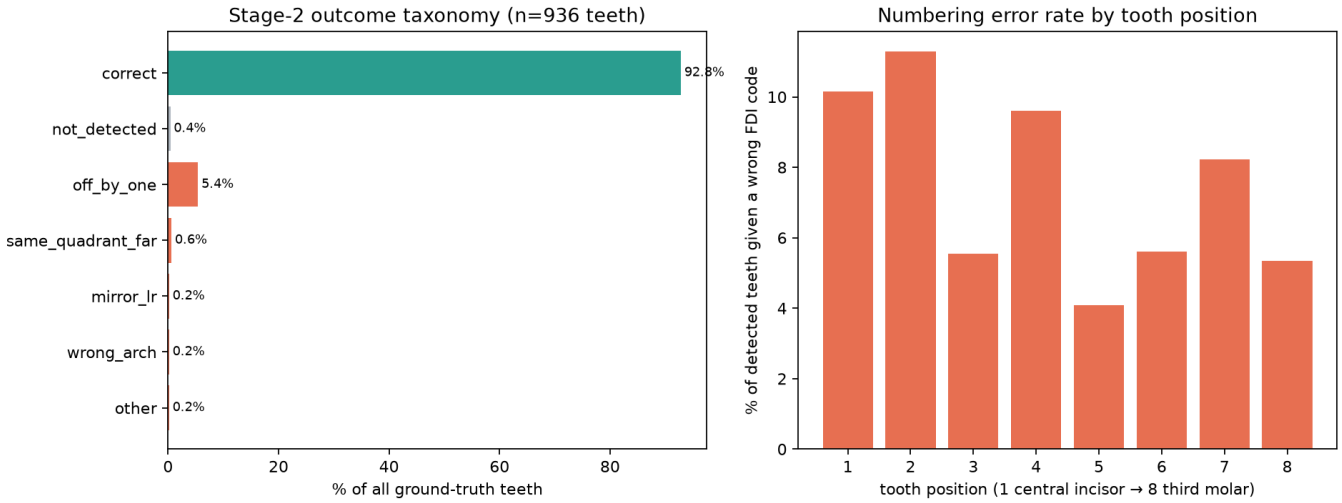


Fig. 2. Where the numbering stage loses teeth, over 936 ground-truth teeth in 164 DENTEX images, in-domain for the tooth detector. Almost every tooth is proposed and numbered correctly; among the codes that are wrong, most are off-by-one within the correct quadrant.

(recall 66.6 %), on the square export as distributed, and 77.2 % [75.19, 79.14] (recall 95.6 %) with each image widened to a panoramic shape, an aspect ratio we have not verified for these files. That is one corpus and a single pass; on it the adopted detector leads the one it replaced by +23.86 points [+21.96, +25.77].

Residual errors on DENTEX are mostly near misses: of 63 incorrect codes, 51 (81.0 %) name an adjacent tooth within the correct quadrant, and only 6 of 932 detected teeth (0.6 %) receive an error at the quadrant level—naming a tooth in the wrong corner of the mouth. A constant cannot attempt this task at all.

Calibration. *Calibration* asks whether a model's confidence behaves like a probability: of the teeth it calls carious with 80 % confidence, are 80 % actually carious? **Expected calibration error** (ECE) is the average gap between stated confidence and observed frequency across confidence bins; 0 is perfect.

Everything in this subsection that concerns caries was measured on the **per-crop gate**, which since Section VI-C is the retained Stage-3a variant rather than the default. We state the consequence rather than leave it implied: the adopted lesion detector emits a detection score, not a posterior over a tooth, so it has no reliability diagram, no calibrator and no abstention curve of its own, and we do not claim one for it (Section VIII). The numbering results below are unaffected—Stage 2 did not change—and they are what the two capabilities look like in the system as it ships today.

The caries gate's confidences are not probabilities. Its ECE is **0.136** and every reliability bin lies below the diagonal—it is systematically overconfident. Temperature scaling [26], the standard first remedy, made calibration *worse* (0.152), indicating the miscalibration is structural rather than a single scale factor. Isotonic regression [27] reduced ECE to **0.043**.

That improvement is not free, and we report its cost. Isotonic regression is only weakly monotonic: on this model it collapses

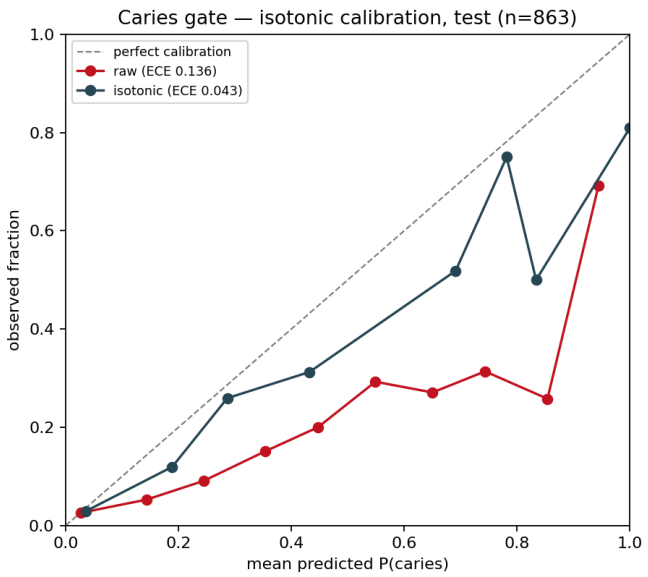


Fig. 3. Reliability of the caries *gate* — the retained Stage-3a variant, not the adopted lesion detector — before and after isotonic regression, on 863 held-out test crops. The uncorrected model is systematically overconfident: at a stated confidence of 0.85 the observed frequency of caries is 0.26.

the test scores to 46 distinct values and average precision falls from **0.550 to 0.500**. Applied at the decision threshold it would also change **155 of 863 verdicts (18 %)**. We therefore apply it to the *reported confidence only*, leaving each verdict as the raw `argmax`, so that every published caries number in this paper is the one that was measured. This is a deployment choice with a stated price, not a property of the method—and it is the opposite of temperature scaling, which cannot change a verdict but did not fix the calibration.

One number should accompany all of this: the gate's average

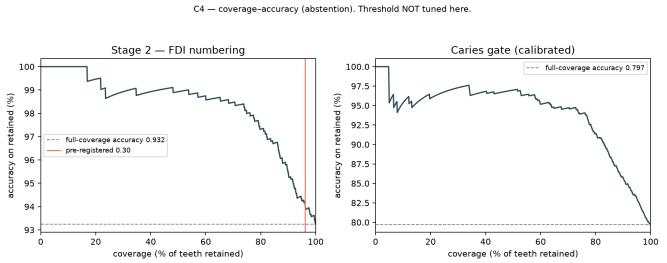


Fig. 4. Coverage–accuracy for tooth numbering (left) and the caries *gate* (right); the caries panel is the retained Stage-3a variant, not the adopted lesion detector. Accuracy on retained predictions rises as the least-confident cases are declined. The vertical line marks the pre-registered threshold of 0.30, located on the curve rather than tuned against it.

precision is **0.550** $[0.445, 0.660]$ against a prevalence of 0.147, i.e. 3.7× chance. What fails is calibration and the operating point, not the model's ability to rank teeth.

Selective prediction. A system may be allowed to *decline*—to answer "I am not confident enough on this tooth" instead of guessing. *Coverage* is the fraction of cases it still answers; accuracy is then measured on those alone. This is only useful if the model's confidence actually identifies its own errors.

Declining the least-confident quarter of cases raises numbering accuracy from 0.932 to **0.980** (+4.76 pp, CI $[+3.29, +6.54]$, 164 image clusters) and caries-gate accuracy from 0.797 to **0.940** (+14.25 pp, CI $[+11.32, +16.63]$, 34 image clusters). Under the shuffle control both gains collapse to −0.06 and +0.00 percentage points, establishing that the improvement comes from the confidence ranking and not from discarding items.

The pre-registered threshold of 0.30 lands at 96.1 % coverage and 0.941 accuracy. It was chosen before these data were seen, and on this corpus it declines little: at 90 % coverage the empirical accuracy is 0.955.

One caveat must accompany any coverage figure here: coverage is measured over *detected* teeth. Retaining 75 % of detections corresponds to 75 % of 99.6 %, or **74.7 % of the teeth actually present in the mouth**, on a corpus where nearly every tooth is found. Where the detector finds fewer, the same sentence has to be recomputed.

E. Vision–language model behaviour

Few-shot prompting, with exemplars drawn strictly from leakage-excluded images, degrades the smaller model monotonically: F1 falls from 0.149 zero-shot to 0.066 at two shots and 0.034 at four. The mechanism is abstention rather than error—empty answers rise from 56 of 164 images to 123 of 164—so exact-set match simultaneously *rises*, because 16 images contain no caries and an empty answer scores an exact match on each. Reported alone, that column would be true and thoroughly misleading.

The content-free controls are more direct, and the two models behave differently enough that they must be reported separately.

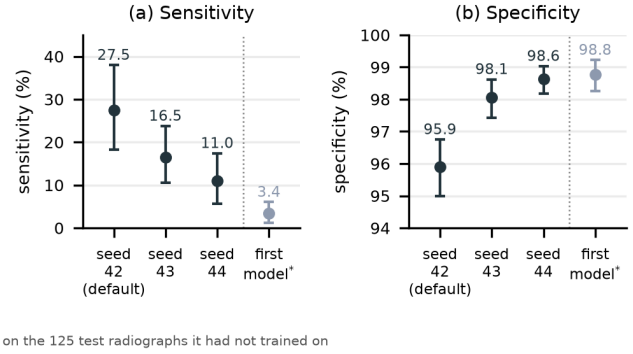


Fig. 5. Stage 4 on the held-out Tufts test split (143 radiographs, 3,892 teeth, 182 carrying its label): three training runs of one recipe, each with a 95 % interval from a bootstrap over radiographs, beside the first model on the 125 of those radiographs it had not trained on. The intervals cover the sampling of radiographs, not the training run; the spread between runs is the larger uncertainty. The axis in (b) is truncated.

- **Qwen2.5-VL-7B-Instruct** answers `26,36` to a blank grey image and `26,36` to random noise—identical to its answer for a real radiograph.
- **Qwen2.5-VL-32B-Instruct** answers `NONE` to the blank image, which is correct behaviour. Shown random noise, it answers `16, 17, 26, 27, 36, 37, 46, 47`—*exactly* the eight-molar set our greedy search recovered from the data, and identical to its answer for a real radiograph.

The larger model therefore detects the absence of an image but not the absence of anatomy, and what it emits in that condition is the population prior itself. Whatever these outputs measure on a noise input, it is not perception of the image in front of them. We claim this for these two models under this prompt, and no further.

The efficiency comparison survives all of the above: 33.45 B parameters and 70.1 GB of accelerator memory are outscored by 72.99 M parameters running on a CPU—a factor of roughly 460 in parameter count, in the wrong direction for the larger model.

F. Stage 4: an abnormality flag, measured held out

Stage 4 was built to flag periodontal bone loss, and it is not a periodontal detector. It is a ResNet18 that flags teeth whose crop overlaps an expert-marked abnormality on Tufts—described by that expert as apical lesions, sclerosis and cysts—so we name it for that label. The model first shipped reported 0.923 macro-F1 and 83 % sensitivity from a crop-level split in which every validation radiograph also supplied training crops; on the 125 test radiographs it had not trained on, it found **3.4 %** of such teeth (6 of 175). Retrained with the split made by radiograph, under a protocol written down before training, three runs of one recipe found **27.5 %, 16.5 % and 11.0 %** of such teeth on 143 test radiographs (50, 30 and 20 of 182; 18.3 % pooled), at 95.9–98.6 % specificity, with about seven flags in ten false (precision 24.8–29.4 %; Fig. 5). The runs rank teeth alike (ROC area 0.68–0.69) and differ in operating

point; the ranking is separated from a tooth-position prior on average precision and not on ROC area. The run whose seed was fixed in advance, the 27.5 % one, is the pipeline default; that choice was made with the test result in view, and we do not call it pre-registered.

These figures were scored on annotated tooth boxes, not on the pipeline's own detections, and the flag has no external figure: on the 164 DENTEX images, which carry no such label, it fires on 6.1 % of detected teeth (274 of 4,512), a rate and not a sensitivity. The three runs share one pre-computed set of jittered training crops, so their spread understates training variance, and every interval covers the sampling of radiographs only. One of the 143 test radiographs is a re-rendered copy of a training radiograph; it holds no positive tooth. A second ResNet18, meant to grade the flagged teeth, was trained on bands of crop contrast, a stand-in for an overlap measurement its training data lacked; it is present in the released code and withheld from the report by default. Retrained on bands of the share of the tooth crop overlapping an expert-marked abnormality—an extent band, not a severity—the head reached macro-F1 0.294 [0.214, 0.385] on 182 held-out positive teeth, against 0.334 for an image-blind tooth-position prior; two further runs gave 0.328, below the prior, and 0.382. It grades only teeth that truly carry the label; in the pipeline most flagged teeth do not. The rule registered before training, that this run beat the prior with a 95 % interval above zero, was not met (difference -0.04, interval $[-0.12, +0.06]$), so the grade stays withheld and no retrained head ships. The contrast-trained head scored 0.366 on the same teeth (0.371 without the three test radiographs it trained on). We make no periodontal claim.

VII. Discussion

A. Main finding

On our development data an image-blind constant outscored two vision–language models, and a purpose-built six-stage pipeline cleared it—by four points with one caries stage and eighteen with another—under a tooth detector trained on that corpus. On four external hospitals the pipeline with its per-crop stage and the constant could not be separated at all, and on a second external benchmark the constant led by 22 points. The finding is therefore not that the constant wins—it does not, reliably—nor that the pipeline does, but that **these metrics cannot tell the two apart outside the data a system was built on**, which is a stronger and more troubling result than either outcome alone.

That conclusion is about a task, and it survives being acted on. Reformulating Stage 3a from per-tooth classification to lesion detection—the same backbone family, no supervision beyond boxes the corpora already carried—moved the external comparison from a tie to a 9.28-point lead over the constant, on the same 337 images and the same metric (Section VI-C). What changed was the question the model was asked, not how well it was asked. We read the two results as one finding rather than two: a tooth-level agreement figure is uninformative about perception, and the way past that is to score something the prior cannot produce—not to build a better classifier of the same thing.

B. Why the floor is high, and why that is not a criticism of any model

Caries in these cohorts is anatomically concentrated: molars carry most of the disease. A tooth-level decision is a set-membership question over a fixed vocabulary of 32 codes, and set membership inherits that concentration directly. A fixed answer naming the eight first and second molars is not a degenerate output—it is a compact statement of dental epidemiology, and it is right for the same reason a clinician's prior is right.

The consequence is that the headroom above such an answer is small, and small *independently of model quality*. No amount of architectural work raises a ceiling imposed by the pairing of task and metric. A system distinguishes itself here only by being right about *this patient* in a way the population prior is not, and tooth-level set membership is a poor instrument for detecting that difference.

We treated that as a claim to be tested rather than a rhetorical flourish, and it held in both directions. Five separate repairs to the classifier returned intervals spanning zero; the ceiling moved only once the pairing itself did (Section VI-C).

The floor also moves with the front end, which is the detail we did not anticipate. Changing only the tooth detector raised the per-crop pipeline's lesion agreement on every cohort—and raised the constant's with it, by $+13.99$, $+4.12$ and $+19.25$ points on the three cohorts, because the constant's eight codes are placed on the image by the pipeline's own boxes and numbering. An improvement of that kind is real and worth having. It is not evidence that the caries stage reads lesions any better, and only the comparison against the constant, which received the same improvement, can tell the two apart.

C. Why the external disagreement changes the interpretation

We expected the external cohort to confirm the development result. It did not. Pooled, the two systems are indistinguishable; per site, the difference swings through 31 points.

Two readings remain open and we cannot separate them with the data we have. Either the development lead is partly a property of DENTEX—of its particular caries distribution, or of the tooth detector having trained on it—or the external metric, which must use centroid containment for the geometric reason given in Section V, has different sensitivity to the same underlying effect. We report the ambiguity rather than resolve it by assertion.

What the disagreement does establish is firmer than what it leaves open: at this variance, *a single external number carries very little information*. Our own pooled 0.600 is, in isolation, as uninterpretable as any other pooled figure in this literature, including the ones we cite. One hospital in our cohort would have supported the claim that the constant wins decisively; another would have supported the opposite. Reporting either alone would have been defensible, publishable, and wrong.

D. What the system can still demonstrate

A constant cannot name a tooth, cannot express a calibrated confidence, cannot decline informatively, and has no behaviour under domain shift to measure. We propose these as the axes on which OPG systems should be compared, with our own figures as an illustration of what reporting them looks like:

1) **Identification**—can it say *which* tooth? Here 93.2 % of detected teeth on one in-domain corpus and 71.1 % on another, with detection recall of 99.6 % and 89.3 % stated alongside.
2) **Calibration**—do its confidences behave like probabilities? For numbering, yes; for caries, ECE $0.136 \rightarrow 0.043$ *on the gate variant*, with the failed temperature-scaling attempt reported because it diagnoses the miscalibration as structural, and with the $0.550 \rightarrow 0.500$ average-precision cost stated.
3) **Selective prediction**—does declining improve accuracy, and does the gain survive a shuffle control? $+4.76$ pp for numbering and $+14.25$ pp for the caries gate, and yes to the control in both cases.
4) **Domain shift**—measured on held-out institutions and reported per site, including when the answer is that a branch does not transfer.

A constant scores nothing on any of these. That asymmetry, rather than a caries F1, is what the system has to show. We hold ourselves to it: on axes 2 and 3 the caries branch as it now ships has no result, because the stage those numbers describe is no longer the default one (Section VIII).

E. Anticipated objections

"Your pipeline is simply weak." Possibly, but that does not explain the result. Two general-purpose models at 8.29 B and 33.45 B parameters were scored by the identical scorer and placed *below* the constant, and our pipeline, which clears it on development data, could not be separated from it at four hospitals it had never seen. An explanation resting on our system's quality would have to account for three systems of very different design failing to separate from the same prior once the home advantage is removed.

"The constant was fitted, so it is not uninformed." It is fitted—on 78 held-out images, byte-disjoint from every scored image—and then frozen. What it learns is a population statistic: which teeth tend to be carious, not anything about the radiograph in front of it. That is exactly the quantity a diagnostic claim must exceed. Its score drops from 0.392 on the fitting images to 0.320 on the scored ones, in the direction an honest held-out fit should.

"The metrics were poorly chosen." The first is the metric this literature predominantly reports; we did not select it to make a point. The second we introduced specifically to escape the first, and outside the development corpus it gave the same answer. We also report the degenerate flag-everything arm, which reaches 0.998 agreement, so the reader can judge the metric's headroom directly.

"A molar-prior system would still be clinically useful." It might well be, and we do not dispute it. But a tool that encodes epidemiology is a different object from a tool that reads a radiograph, and only the second can find the lesion the prior does not expect. That distinction is invisible to the metric and central to the clinic.

F. What we propose should be reported

An image-blind baseline row. One line of code, no compute, alongside any tooth-level caries metric on this class of data. Without it a mid-range number cannot be interpreted—the five rows of Table IV span 0.149 to 0.498, and the one that reads no image sits in the middle of them.

Per-institution results. Pooled external figures conceal exactly the variation that external validation exists to expose.

Metrics with demonstrated headroom. If a degenerate arm reaches most of the achievable range, the metric cannot support a perception claim regardless of what a system scores on it.

None of this invalidates prior work. It narrows one specific inference: the claim *this system perceives caries* is not established by a tooth-level number alone.

G. Clinical interpretation

This study does not establish clinical usefulness, and nothing here should be read as a claim about patient outcomes. We did not compare any system against a dentist, did not evaluate prospectively, and did not measure whether a flagged tooth changes what a clinician does. What the study does support is a narrower statement about how such systems should be assessed before they reach a clinic: that a tooth-level agreement score, reported without a floor and without external per-site results, is not sufficient evidence that a screening aid is reading the image it is shown.

H. On reporting results that do not favour the authors

Three findings here are unfavourable to our own system: an external comparison in which it cannot be told from a constant, a fourth stage whose first score came from a leaky split and whose held-out recall is low, and a tooth detector adopted over a registered guard it failed. We report all three. A screening tool that fails silently is more dangerous than one that fails visibly, and the alternative—selecting whichever metric flatters the system—is precisely the practice this paper argues against. We would have no standing to recommend image-blind baselines to others while declining to survive one ourselves.

VIII. Limitations

Each item states what it prevents us from concluding.

1) **The adopted caries stage has no calibrated confidence and no abstention curve.** The calibration and selective-prediction results of Section VI-D were measured on the per-crop gate, which Section VI-C demoted to a retained variant. The lesion detector that replaced it emits a detection score, and we have neither fitted a calibrator for it nor re-derived a coverage–accuracy curve against it; the pre-registered 0.30 threshold was

located on the gate's curve and does not transfer to a quantity with a different meaning. So for caries, the system as it ships today demonstrates identification and domain-shift behaviour but neither of the middle two axes of Section VII-D. Numbering demonstrates all four. We regard closing this as the first piece of follow-up work, and it needs its own pre-registration rather than a threshold chosen once the curve is visible.

2) **No clinician comparison.** We do not know how any system here compares against a dentist on these images. We therefore cannot claim clinical utility, non-inferiority, or that the constant would mislead a clinician less than the pipeline. This is the single largest gap.
3) **The caries cohorts disagree in direction.** The pipeline with its per-crop stage leads the constant on development data, ties it across four external hospitals and trails it on a second external benchmark. We cannot claim that either wins in general; the supported claim is non-separability outside the development corpus.
4) **Internal and external caries figures are not comparable to each other.** They use different lesion matchers, forced by a 200-fold difference in lesion-to-tooth area. No internal-versus-external degradation figure should be read off Tables V and VI.
5) **Tooth-level is our finest localisation.** The localisation metric is a tooth-level correspondence check, not a pixel-level segmentation assessment. We cannot claim the system localises *within* a tooth.
6) **The external cohort lacks healthy controls.** Specificity, AUC and false-positive rate are not computable there; we report agreement only, which cannot be penalised for over-flagging.
7) **Unmatched predicted-positive teeth are not confirmed false positives.** DENTEX annotates only teeth carrying some finding—roughly 6 of about 30 per image—so most teeth are unannotated rather than verified healthy, and not one tooth in the 164 scored images is labelled healthy, so no false-positive rate can be given for any caries arm. The match rate is reported as a diagnostic, never as precision.
8) **One VLM family at two scales.** A domain-adapted medical VLM was not accessible. We cannot generalise the control result to VLMs as a class, only to these two models under this prompt.
9) **Numbering is strong only where the tooth detector was trained.** Both corpora that carry tooth labels contributed training images to it, so 92.8 % on DENTEX and 63.5 % on Tufts are held-out but in-domain figures, 29 points apart; on UFBA-425, the one corpus neither model trained on, it is 44.5 % as distributed (Section VI-D). On the Tufts test split the detector scores 0.587 box mAP@0.5, 13.6 points below a Tufts-only control, and it was adopted over a registered guard on that figure (Section IV). All coverage figures are fractions of the teeth it detects.
10) **Stage 4's label is not periodontal**, and its figures come from one institution, three training runs that share their augmented crops, and annotated tooth boxes rather than the pipeline's own detections (Section VI-F). Its severity head was trained on crop contrast and is withheld; retrained on the extent band it was named for, it did not beat an image-blind tooth-position prior.
11) **Every model is one training run**, except Stage 4's three. Every interval resamples images or radiographs, not training runs, so none of them covers training variance.
12) **Patient-level separation cannot be verified on DENTEX**, which ships no patient identifiers. We establish byte-level and image-level separation only.
13) **All data is retrospective and public or archival.** There is no prospective evaluation, and prevalence in these cohorts need not match any clinic's.

IX. Conclusion

We evaluated an automated screening pipeline for dental caries on panoramic radiographs, with a fourth stage that flags teeth near expert-marked abnormalities, and measured what its scores mean by comparing it against progressively less informed systems under one scorer.

On development data, a fixed list of eight molar codes—emitted for every radiograph, fitted on held-out images and frozen—outscored two vision–language models, and our six-stage pipeline exceeded it only with a tooth detector trained on that same corpus. A tooth-level score in this range therefore cannot on its own establish that a system is using image information. Across four external hospitals the pipeline's per-crop caries stage and the constant could not be separated at all, with a 31-point swing between sites that the pooled figure erases, and on a second external benchmark the constant led by 22 points; a single pooled external number carries correspondingly little information. Both language models returned confident, correctly formatted, anatomically specific answers to an image containing pure noise, and the larger one's answer in that condition was precisely the population prior. Recasting the caries stage as lesion detection is the one change that moved the external comparison: 9.28 points above the constant on the four hospitals pooled, while flagging fewer teeth than it. The fourth stage's first score, 0.923 macro-F1, came from a split that let the same radiographs into training and validation; split by radiograph, it found 11.0–27.5 % of its label across three training runs.

Four capabilities distinguished the system from the prior where the caries score could not, and the numbering stage shows all four: identifying which tooth, producing confidences that behave like probabilities, declining informatively with a gain that survives a shuffle control, and being measurable under domain shift. For caries, the middle two were shown on the retained per-crop gate, not on the lesion detector that ships (Section VIII). These cost little to report.

We therefore recommend three things of the next paper reporting a tooth-level caries result: publish an image-blind baseline beside it, report external results per institution rather

than pooled, and state the headroom of the metric being used. We do not conclude that automated caries screening does not work. We conclude that a tooth-level classification score, on its own, cannot tell us whether it does, and that asking where the lesions are is the more informative test.

Reproducibility

Every figure is rendered from the saved evaluation artefacts at build time rather than transcribed. All model checkpoints are SHA-256 pinned, and each evaluation artefact records the hashes of the weights that produced it. The frozen constant reported in Table IV was independently re-derived from the 78 fitting images during manuscript preparation and reproduced exactly. Exact package versions are recorded in `requirements-lock.txt`. The supporting test suite comprises 194 tests, all passing.

References

[1] Y.-S. Lim, D. Chun, J. Kim, J.-y. Lee, M. J. Ju, J. H. Park, and H.-J. Jeon, "Dual convolutional neural network framework for segmenting dental caries in panoramic radiographs," *Journal of Prosthetic Dentistry*, vol. 135, pp. 403.e1–403.e8, 2026. [Online]. Available: https://doi.org/10.1016/j.prosdent.2025.09.040

[2] B. Chen, S. Fu, Y. Liu, J. Pan, G. Lu, and Z. Zhang, "CariesXrays: Enhancing caries detection in hospital-scale panoramic dental X-rays via feature pyramid contrastive learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 20. AAAI Press, 2024, pp. 21 940–21 948. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/30196

[3] A. Bilal, A. H. Khan, K. Almohammadi, S. A. Al Ghamdi, H. Long, and H. Malik, "PDCNET: Deep convolutional neural network for classification of periodontal disease using dental radiographs," *IEEE Access*, vol. 12, pp. 150 147–150 168, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10400831

[4] Ö. Mutlu, E. Aslan, and A. Mert, "Deep learning–based automated diagnostic charting on panoramic radiography: Comparison of YOLOv11 and YOLOv12," *Odontology*, 2026.

[5] K. A. Asif and R. A. Khan, "Dental panoramic radiograph analysis using YOLO26 from tooth detection to disease diagnosis," *arXiv preprint arXiv:2604.16231*, 2026. [Online]. Available: https://arxiv.org/abs/2604.16231

[6] K. Nassiri and M. A. Akhloufi, "YOLO-based panoramic dental X-ray image analysis," *Neural Computing and Applications*, vol. 37, pp. 25 867–25 890, 2025. [Online]. Available: https://link.springer.com/article/10.1007/s00521-025-11462-5

[7] S. Eftimie, T. Ileni, L. Iacob, M. Hedesiu, and L. Diosan, "Tooth-level detection and mapping of dental pathologies on panoramic radiographs using YOLOv11 and RT-DETR," *MethodsX*, vol. 15, no. 3, p. 103696, 2025, published as Sorana Eftimie; an earlier SSRN preprint lists the same author as Sorana Muresanu. Distinct from the unrelated Mureşanu 2024 *Diagnostics* paper already in this corpus (`muresanu2024automating`). [Online]. Available: https://doi.org/10.1016/j.mex.2025.103696

[8] I. D. S. Chen, C.-M. Yang, M.-J. Chen, M.-C. Chen, R.-M. Weng, and C.-H. Yeh, "Deep learning-based recognition of periodontitis and dental caries in dental X-ray images," *Bioengineering*, vol. 10, no. 8, p. 911, 2023. [Online]. Available: https://www.mdpi.com/2306-5354/10/8/911

[9] S. S. Alharbi, A. A. AlRugaibah, H. F. Alhasson, and R. U. Khan, "Detection of cavities from dental panoramic X-ray images using nested U-Net models," *Applied Sciences*, vol. 13, no. 23, p. 12771, 2023. [Online]. Available: https://www.mdpi.com/2076-3417/13/23/12771

[10] B. G. Zancan, J. A. Carneiro, C. U. Martins, C. Tirapelli, C. P. Capel, E. D. da Costa, H. Gaêta-Araujo, J. A. Baranauskas, and A. A. Macedo, "AI-powered precision in dental radiographic analysis using tailored CNNs for tooth numbering and cavity detection," *PLOS Digital Health*, vol. 4, no. 2, p. e0001074, 2025. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC11868773/

[11] Y. Zhang, Y. Yu, Q. Li, H. Chen, Y. Liang, J. Chen, and J. Yin, "A reliable multi-stage system for tooth instance segmentation and numbering in panoramic radiographs," *IEEE Journal of Biomedical and Health Informatics*, pp. 1–13, 2025, early access. [Online]. Available: https://doi.org/10.1109/JBHI.2025.3643805

[12] Z. Meng, J. Hao, X. Dai, Y. Feng, J. Liu, B. Feng, H. Wu, X. Gai, H. Zhu, T. Hu, Y. Wu, H. Xu, J. Li, J. Xiao, X. Liu, J. T. Zhou, F. Zhu, Z. Zhao, L. Xia, B. Fang, J. Sun, J. Wu, and Z. Liu, "A multimodal vision-language model for comprehensive dental diagnosis and enhanced clinical practice," *Nature Communications*, vol. 17, p. 8933, 2026, article 8933. Metadata verified against the Crossref record, 28 Aug 2026. Preprint precursor available as arXiv:2509.23344 (DentVLM). [Online]. Available: https://doi.org/10.1038/s41467-026-75718-x

[13] D. Jha, T. K. Dutta, R. Paudel, O. K. Susladkar, A. Ghimire, A. Dhakal, S. Adhikari, N. K. Yadav, D. R. Nayak, P. Pawar, W. C. W. Chen, and G. Reynolds, "DentiAsk: A VQA benchmark for multimodal reasoning in panoramic dental radiographs," *arXiv preprint arXiv:2607.08790*, 2026. [Online]. Available: https://arxiv.org/abs/2607.08790

[14] X. Li, X. Liu, C. Liu, S. W. Au Yeung, J. Liu, Y. Yuan, and H. Chen, "PanDent: Toward comprehensive tooth-level structure-language consistency in dental radiology," *arXiv preprint arXiv:2607.27378*, 2026. [Online]. Available: https://arxiv.org/abs/2607.27378

[15] J. Hao, Y. Fan, Y. Sun, K. Guo, L. Lin, J. Yang, Q. Y. H. Ai, L. M. Wong, H. Tang, and K. F. Hung, "Towards better dental AI: A multimodal benchmark and instruction dataset for panoramic x-ray analysis," in *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*. Curran Associates, 2025, also available as arXiv:2509.09254. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/file/3797b3a3c939156757857cf1b9afde3c-Paper-Datasets_and_Benchmarks_Track.pdf

[16] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.

[17] Z. Cao, C. Feng, Y. Shen, G. Ye, H. Xu, J. Wu, Z. Wu, H. Gao, and H. Zhu, "DPML: Prior-guided multitask learning for dental object recognition on limited panoramic radiograph dataset," *Expert Systems with Applications*, vol. 254, p. 124446, 2024.

[18] D. Cerda Mardini, P. Cerda Mardini, D. Paz Vicuña Iturriaga, and D. R. Ortuño Borroto, "Determining the efficacy of a machine learning model for measuring periodontal bone loss," *BMC Oral Health*, vol. 24, no. 1, p. 100, 2024. [Online]. Available: https://bmcoralhealth.biomedcentral.com/articles/10.1186/s12903-023-03819-w

[19] G. Jocher, J. Qiu, and A. Chaurasia, "Ultralytics YOLO11," Software, version 8.3, 2024, https://github.com/ultralytics/ultralytics.

[20] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[21] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.

[22] K. Panetta, R. Rajendran, A. Ramesh, S. P. Rao, and S. Agaian, "Tufts dental database: A multimodal panoramic x-ray dataset for benchmarking diagnostic systems," *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 4, pp. 1650–1659, 2022.

[23] I. E. Hamamci, S. Er, E. Simsar, A. Sekuboyina, and Others, "DENTEX: An abnormal tooth detection with dental enumeration and diagnosis benchmark for panoramic x-rays," 2023, arXiv:2305.19112.

[24] X. Li, X. Ma, Y. Zhao, J. Hu, J. Liu, Z. Yang, F. Han, J. Zhang, W. Liu, and Z. Zhou, "A multi-center dental panoramic radiography image dataset for impacted teeth, periodontitis, and dental caries: Benchmarking segmentation and classification tasks," *Journal of Imaging Informatics in Medicine*, vol. 37, no. 2, pp. 831–841, 2024.

[25] S. Bai, K. Chen, X. Liu, J. Wang, and Others, "Qwen2.5-VL technical report," 2025, arXiv:2502.13923.

[26] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 70, 2017, pp. 1321–1330. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html

[27] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002, pp. 694–699.